```
### Imports
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import MultipleLocator
from scipy.signal import find_peaks
from scipy.optimize import curve_fit
```

```
def plot_xrd(data, title="XRD Pattern"):
        fig, ax = plt.subplots(figsize=(9, 4.8), dpi=150)

        # plot the XRD curve
        ax.plot(
                data["Angle"],
                data["Intensity"],
                color="blue",
                linewidth=0.6,
                antialiased=True
        )

        # x-axis settings
        xmin = np.floor(data["Angle"].min())    # minimum angle shown on x-axis
        xmax = np.ceil(data["Angle"].max())      # maximum angle shown on x-axis
        ax.set_xlim(xmin, xmax * 1.01)
        ax.set_xlabel("2Theta (deg)")

        major_step = 10
        xticks = np.arange(
                np.ceil(xmin / major_step) * major_step,
                xmax + major_step,
                major_step
        )
        ax.set_xticks(xticks)
        ax.set_xticklabels([""] + [str(int(x)) for x in xticks[1:]])
        ax.xaxis.set_minor_locator(MultipleLocator(1))

        # y-axis settings
        ymax = data["Intensity"].max()
        ax.set_ylim(0, ymax * 1.08)
        ax.set_ylabel("Counts")
        ax.yaxis.set_minor_locator(MultipleLocator(500))

        # tick style
        ax.tick_params(axis="both", which="major",
                                direction="in", length=6, width=0.8)
        ax.tick_params(axis="both", which="minor",
                                direction="in", length=3, width=0.6)

        # frame style
        for spine in ax.spines.values():
                spine.set_visible(True)
                spine.set_linewidth(0.8)

        ax.grid(False)
        ax.set_title(title)

        plt.tight_layout()
        plt.show()
```

```
### Load XRD data
def load_xrd(filename, skiprows=27):
        df = pd.read_csv(
                filename,
                skiprows=skiprows,
                names=["Angle", "Intensity"],
                sep=r"[,\s]+",
                engine="python"
        )

        df["Angle"] = pd.to_numeric(df["Angle"], errors="coerce")
        df["Intensity"] = pd.to_numeric(df["Intensity"], errors="coerce")
        df = df.dropna(subset=["Angle", "Intensity"])
        df = df[np.isfinite(df["Angle"]) & np.isfinite(df["Intensity"])]

        return df.reset_index(drop=True)
```

```
### Linear baseline correction
def linear_baseline_correction(df, x_ranges=((30, 34), (80, 110)), shift_to_zero=False):
        x = df["Angle"].to_numpy()
        y = df["Intensity"].to_numpy()

        mask = np.zeros_like(x, dtype=bool)
        for x1, x2 in x_ranges:
                mask |= (x >= x1) & (x <= x2)

        x_bg = x[mask]
        y_bg = y[mask]

        if x_bg.size < 2:
                raise ValueError("Not enough points selected for baseline fitting.")

        # fit a linear baseline
        m, b = np.polyfit(x_bg, y_bg, 1)
```

```
        baseline = m * x + b
        y_corr = y - baseline

        # shift the corrected curve so the minimum is zero(if needed)
        if shift_to_zero:
                y_corr = y_corr - np.nanmin(y_corr)

        out = df.copy()
        out["Intensity"] = y_corr
        out["Baseline"] = baseline
        return out
```

```
### Raw data
data_seed = load_xrd("IONF-seed.csv")
data_F1     = load_xrd("IONF-F1.csv")
data_F2     = load_xrd("IONF-F2.csv")
data_F3     = load_xrd("IONF-F3.csv")
data_F4     = load_xrd("IONF-F4.csv")
data_F5     = load_xrd("IONF-F5.csv")

### Baseline correction
# sloping background is removed
data_seed_corr = linear_baseline_correction(data_seed, x_ranges=((30, 34), (80, 110)), shift_to_zero=False)
data_F1_corr     = linear_baseline_correction(data_F1,     x_ranges=((30, 34), (80, 110)), shift_to_zero=False)
data_F2_corr     = linear_baseline_correction(data_F2,     x_ranges=((30, 34), (80, 110)), shift_to_zero=False)
data_F3_corr     = linear_baseline_correction(data_F3,     x_ranges=((30, 34), (80, 110)), shift_to_zero=False)
data_F4_corr     = linear_baseline_correction(data_F4,     x_ranges=((30, 34), (80, 110)), shift_to_zero=False)
data_F5_corr     = linear_baseline_correction(data_F5,     x_ranges=((30, 34), (80, 110)), shift_to_zero=False)

data_dict = {
        "seed": data_seed_corr,
        "F1": data_F1_corr,
        "F2": data_F2_corr,
        "F3": data_F3_corr,
        "F4": data_F4_corr,
        "F5": data_F5_corr
}
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

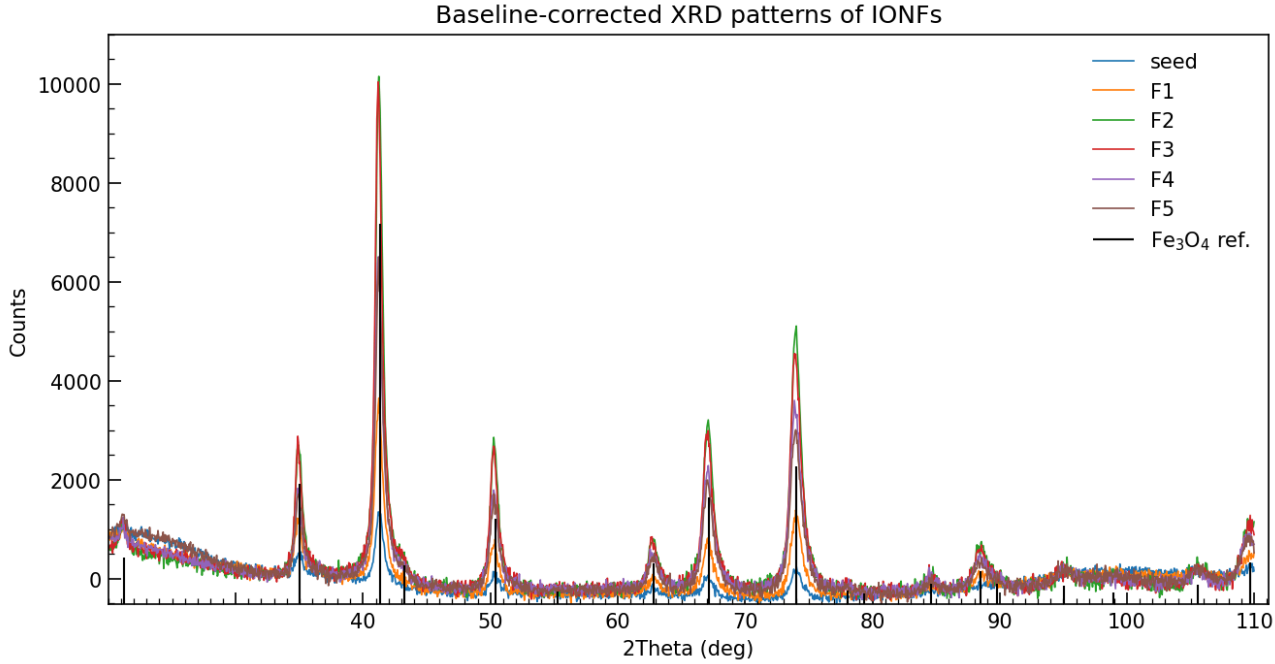
```
### reference peak list
ref_peaks = np.array([
        21.246, 35.039, 41.341, 43.270, 50.393, 55.281,
        62.853, 67.151, 74.036, 78.050, 79.374, 84.619,
        88.521, 89.818, 95.017, 98.941, 105.587, 109.674
])

ref_intensities = np.array([
        9.6, 29.6, 100.0, 7.6, 20.3, 0.6,
        8.1, 26.0, 34.3, 0.8, 0.1, 2.5,
        6.1, 2.5, 2.0, 0.3, 2.3, 8.4
])
```

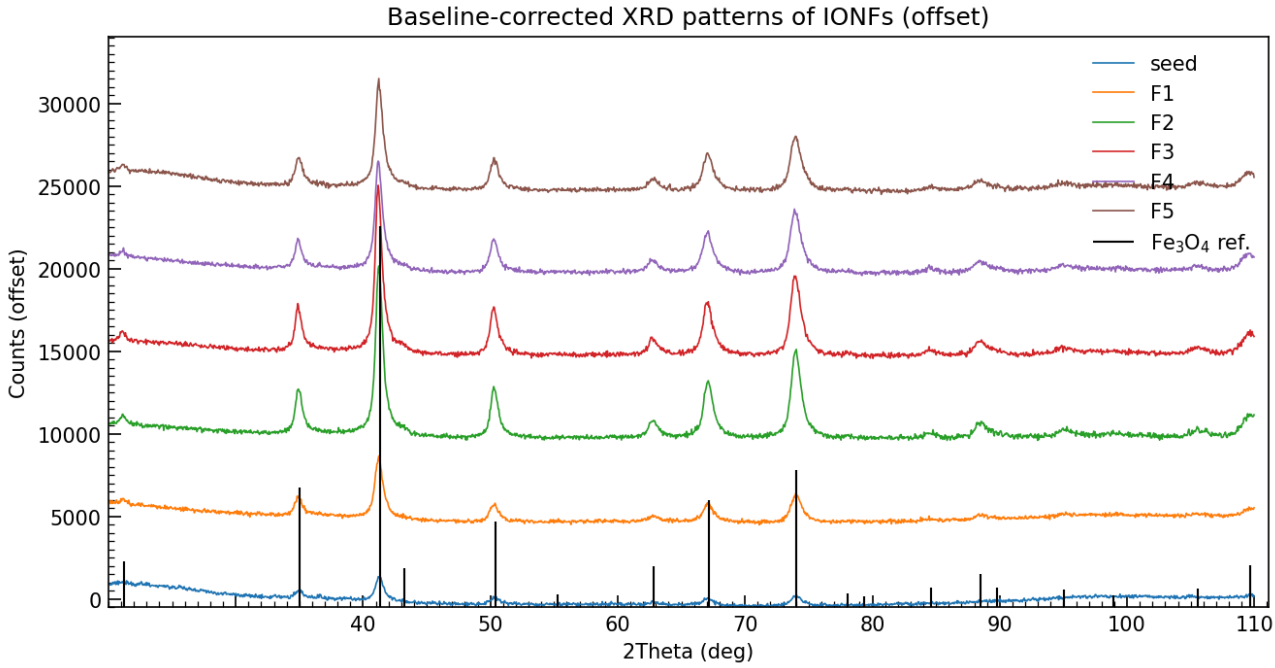
```
### Plot multiple XRD patterns
def plot_xrd_multi(data_dict, title="XRD Patterns", offset=0,
                                ref_peaks=None, ref_intensities=None, ref_label=None,
                                filename=None):

        fig, ax = plt.subplots(figsize=(9, 4.8), dpi=150)

        # Combine all angle values first to determine the x-axis range
        all_angles = []
        for label, df in data_dict.items():
                if df is None or df.empty:
                        continue
                a = df["Angle"].to_numpy()
                a = a[np.isfinite(a)]
                if a.size:
                        all_angles.append(a)

        if len(all_angles) == 0:
                raise ValueError("No valid Angle data found (all files empty or unreadable).")

        all_angles = np.concatenate(all_angles)
        xmin = float(np.nanmin(all_angles))
        xmax = float(np.nanmax(all_angles))

        if not (np.isfinite(xmin) and np.isfinite(xmax)) or xmin >= xmax:
                raise ValueError(f"Invalid x-range: xmin={xmin}, xmax={xmax}")

        ymax_global = 0.0
        ymin_global = np.inf

        for i, (label, data) in enumerate(data_dict.items()):
                if data is None or data.empty:
                        continue

                x = data["Angle"].to_numpy()
                y0 = data["Intensity"].to_numpy()
```

```
            mask = np.isfinite(x) & np.isfinite(y0)
            x = x[mask]
            y0 = y0[mask]

            if x.size == 0:
                continue

            y = y0 + i * offset

            ax.plot(
                x,
                y,
                linewidth=0.8,
                antialiased=True,
                label=label
            )

            ymax_global = max(ymax_global, float(np.nanmax(y)))
            ymin_global = min(ymin_global, float(np.nanmin(y)))

        # Add reference peak sticks if provided
        if ref_peaks is not None and ref_intensities is not None:
            ref_peaks = np.asarray(ref_peaks, dtype=float)
            ref_intensities = np.asarray(ref_intensities, dtype=float)

            mask_ref = (
                np.isfinite(ref_peaks) & np.isfinite(ref_intensities) &
                (ref_peaks >= xmin) & (ref_peaks <= xmax)
            )
            ref_peaks_plot = ref_peaks[mask_ref]
            ref_int_plot = ref_intensities[mask_ref]

            if ref_peaks_plot.size > 0:
                ref_base = min(0.0, ymin_global)
                ref_min_height = (ymax_global - ref_base) * 0.02
                ref_extra_height = (ymax_global - ref_base) * 0.70
                ref_scaled = ref_base + ref_min_height + (ref_int_plot / np.nanmax(ref_int_plot)) * ref_extra_height

                ax.vlines(
                    ref_peaks_plot,
                    ymin=ref_base,
                    ymax=ref_scaled,
                    colors="black",
                    linewidth=1.0,
                    linestyles="-",
                    label=ref_label
                )

        ### X-axis
        ax.set_xlim(np.floor(xmin), np.ceil(xmax) * 1.01)
        ax.set_xlabel("2Theta (deg)")

        major_step = 10
        start = np.ceil(xmin / major_step) * major_step
        xticks = np.arange(start, xmax + major_step, major_step)
        ax.set_xticks(xticks)

        labels = [str(int(x)) for x in xticks]
        if len(labels) > 0:
            labels[0] = ""
        ax.set_xticklabels(labels)

        ax.xaxis.set_minor_locator(MultipleLocator(1))

        ### Y-axis
        if ymax_global <= 0 or not np.isfinite(ymax_global):
            ymax_global = 1.0
        if not np.isfinite(ymin_global):
            ymin_global = 0.0

        ypad = 0.08 * (ymax_global - ymin_global if ymax_global > ymin_global else 1.0)
        ax.set_ylim(ymin_global - 0.02 * abs(ymin_global), ymax_global + ypad)
        ax.set_ylabel("Counts" if offset == 0 else "Counts (offset)")
        ax.yaxis.set_minor_locator(MultipleLocator(500))

        ### Ticks and frame
        ax.tick_params(axis="both", which="major",
                       direction="in", length=6, width=0.8)
        ax.tick_params(axis="both", which="minor",
                       direction="in", length=3, width=0.6)

        for spine in ax.spines.values():
            spine.set_visible(True)
            spine.set_linewidth(0.8)

        ax.grid(False)
        ax.set_title(title)
        ax.legend(frameon=False)

        plt.tight_layout()
        if filename is not None:
            plt.savefig(filename, dpi=300, bbox_inches="tight")
        plt.show()
```

```
### Plot without offset
plot_xrd_multi(
    data_dict
```

```
        data_dict,
        title="Baseline-corrected XRD patterns of IONFs",
        offset=0,
        ref_peaks=ref_peaks,
        ref_intensities=ref_intensities,
        ref_label="Fe$_3$O$_4$ ref.",
        filename="XRD_IONFs_corrected.png"
)

### Plot with offset
plot_xrd_multi(
        data_dict,
        title="Baseline-corrected XRD patterns of IONFs (offset)",
        offset=5000,
        ref_peaks=ref_peaks,
        ref_intensities=ref_intensities,
        ref_label="Fe$_3$O$_4$ ref.",
        filename="XRD_IONFs_corrected_offset.png"
)
```






```
def find_xrd_peaks(data, sample_name, prominence=800, distance=5):
```

```
        x = data["Angle"].values
        y = data["Intensity"].values

        peaks, _ = find_peaks(y, prominence=prominence, distance=distance)

        peak_angles = x[peaks]
        peak_intensities = y[peaks]

        order = np.argsort(peak_intensities)[::-1]
        peak_angles = peak_angles[order]
        peak_intensities = peak_intensities[order]

        print(f"\nMain peaks for {sample_name}:")
        for a, i in zip(peak_angles, peak_intensities):
            print(f"  2theta = {a:.2f}°, Intensity = {i:.0f}")

        return peak_angles, peak_intensities
```

```
find_xrd_peaks(data_seed_corr, "IONF-seed")
find_xrd_peaks(data_F1_corr,   "IONF-F1")
find_xrd_peaks(data_F2_corr,   "IONF-F2")
find_xrd_peaks(data_F3_corr,   "IONF-F3")
find_xrd_peaks(data_F4_corr,   "IONF-F4")
find_xrd_peaks(data_F5_corr,   "IONF-F5")
```

```
Main peaks for IONF-seed:
  2theta = 41.19°, Intensity = 1345

Main peaks for IONF-F1:
  2theta = 41.24°, Intensity = 3648
  2theta = 73.98°, Intensity = 1384
  2theta = 34.82°, Intensity = 1221
  2theta = 67.06°, Intensity = 816
  2theta = 50.37°, Intensity = 794

Main peaks for IONF-F2:
  2theta = 41.24°, Intensity = 10152
  2theta = 74.03°, Intensity = 5105
  2theta = 67.11°, Intensity = 3207
  2theta = 50.27°, Intensity = 2854
  2theta = 34.92°, Intensity = 2702
  2theta = 62.80°, Intensity = 803
  2theta = 88.57°, Intensity = 747
  2theta = 95.29°, Intensity = 432

Main peaks for IONF-F3:
  2theta = 41.19°, Intensity = 10038
  2theta = 73.88°, Intensity = 4547
  2theta = 67.11°, Intensity = 2990
  2theta = 34.87°, Intensity = 2877
  2theta = 50.27°, Intensity = 2676
  2theta = 62.70°, Intensity = 843
  2theta = 88.57°, Intensity = 677

Main peaks for IONF-F4:
  2theta = 41.19°, Intensity = 6503
  2theta = 73.88°, Intensity = 3600
  2theta = 67.11°, Intensity = 2282
  2theta = 34.87°, Intensity = 1824
  2theta = 50.27°, Intensity = 1788
  2theta = 62.55°, Intensity = 596
  2theta = 88.42°, Intensity = 497

Main peaks for IONF-F5:
  2theta = 41.24°, Intensity = 6500
  2theta = 73.98°, Intensity = 3007
  2theta = 67.06°, Intensity = 1991
  2theta = 34.97°, Intensity = 1714
  2theta = 50.27°, Intensity = 1711
  2theta = 62.90°, Intensity = 515
(array([41.24166603, 73.9790401 , 67.06057514, 34.97494052, 50.26575077,
        62.8994694 ]),
 array([6500.20452534, 3006.58553331, 1990.60608597, 1714.1361854 ,
        1710.74293489,  515.09670825]))
```

```
### Gaussian peak function
def gaussian(x, A, mu, sigma):
        return A * np.exp(-(x - mu)**2 / (2 * sigma**2))
```

```
### Gaussian fit for a single XRD peak
def fwhm_at_peak_gaussian(
        data,
        center,
        fit_window=2.5,
        bg_window=5.0,
        plot=False,
        offset=0,
        label=None,
        ax=None
):
        x = data["Angle"].values
        y = data["Intensity"].values

        mask_bg = (x > center - bg_window) & (x < center + bg_window)
        x_bg = x[mask_bg]
        y_bg = y[mask_bg]
        if len(x_bg) < 50:
```

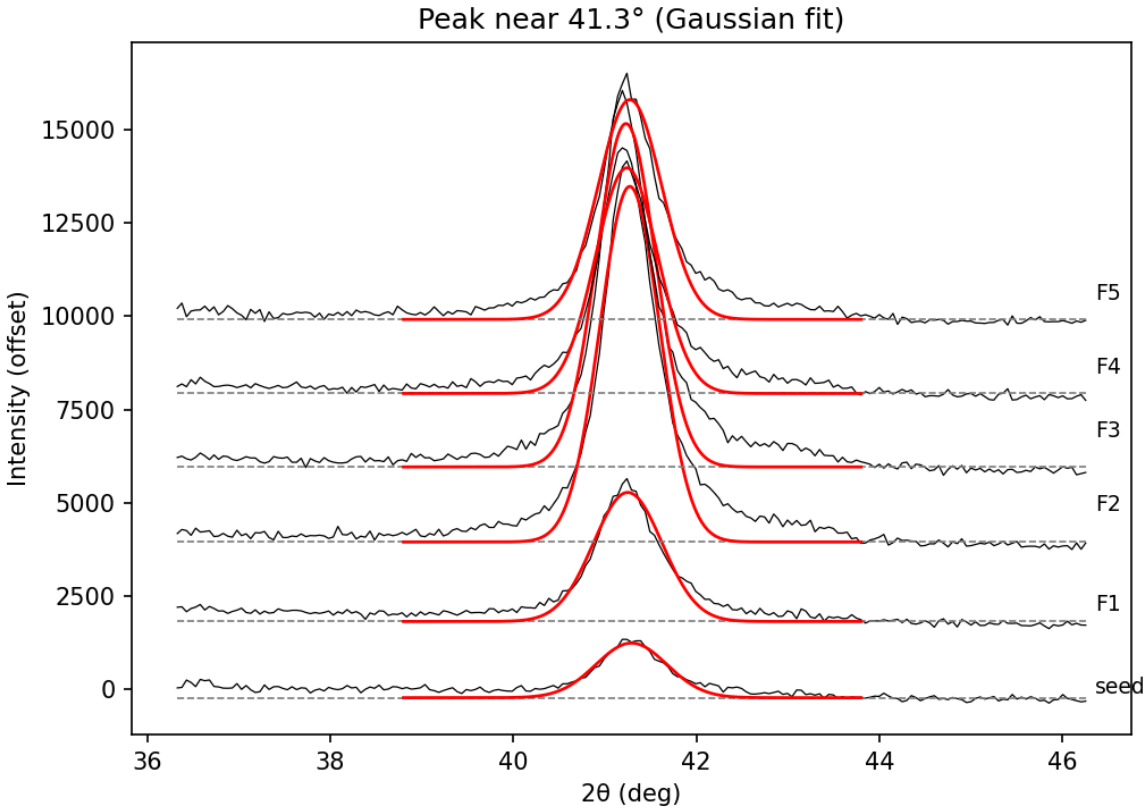
```
        return None

    # estimate the background from the two edges
    n_bg = len(x_bg)
    edge = int(0.15 * n_bg)
    x_bg_edge = np.concatenate([x_bg[:edge], x_bg[-edge:]])
    y_bg_edge = np.concatenate([y_bg[:edge], y_bg[-edge:]])

    b = np.mean(np.sort(y_bg)[:int(0.35 * len(y_bg))])
    k = 0.0

    mask_fit = (x > center - 0.8) & (x < center + 0.8)
    x_win = x[mask_fit]
    y_win = y[mask_fit]
    if len(x_win) < 30:
        return None

    y_corr = y_win - (k * x_win + b)

    A0 = y_corr.max()
    mu0 = x_win[np.argmax(y_corr)]
    sigma0 = 0.15

    try:
        popt, pcov = curve_fit(
            gaussian,
            x_win,
            y_corr,
            p0=[A0, mu0, sigma0],
            bounds=(
                [0, center - 0.8, 0.05],
                [np.inf, center + 0.8, 1.5]
            )
        )
    except RuntimeError:
        return None

    A, mu, sigma = popt
    sigma = abs(sigma)

    perr = np.sqrt(np.diag(pcov))
    A_err, mu_err, sigma_err = perr
    sigma_err = abs(sigma_err)

    # convert sigma to FWHM
    factor = 2 * np.sqrt(2 * np.log(2))
    fwhm_deg = factor * sigma
    fwhm_err_deg = factor * sigma_err

    y_pred = gaussian(x_win, A, mu, sigma)
    ss_res = np.sum((y_corr - y_pred) ** 2)
    ss_tot = np.sum((y_corr - np.mean(y_corr)) ** 2)
    r_squared = 1 - ss_res / ss_tot if ss_tot > 0 else np.nan

    if plot:
        if ax is None:
            fig, ax = plt.subplots(figsize=(7, 4), dpi=150)

        bg_plot = k * x_bg + b
        x_fit = np.linspace(center - fit_window, center + fit_window, 600)
        y_fit = gaussian(x_fit, A, mu, sigma) + (k * x_fit + b)

        ax.plot(x_bg, y_bg + offset, color="black", lw=0.6)
        ax.plot(x_bg, bg_plot + offset, color="gray", ls="--", lw=0.8)
        ax.plot(x_fit, y_fit + offset, color="red", lw=1.3)

        if label is not None:
            ax.text(
                x_bg.max() + 0.1,
                np.mean(y_bg + offset),
                label,
                va="center",
                fontsize=9
            )

        ax.set_xlabel("2θ (deg)")
        ax.set_ylabel("Intensity (offset)")
        ax.set_title(f"Peak near {center}° (Gaussian fit)")

    return mu, fwhm_deg, fwhm_err_deg, r_squared
```

```
### Scherrer size calculation
def scherrer_size(fwhm_deg, peak_deg, fwhm_err_deg=None, K=0.89, wavelength=0.1789):
    beta = np.deg2rad(fwhm_deg)
    theta = np.deg2rad(peak_deg / 2)

    D = K * wavelength / (beta * np.cos(theta))

    if fwhm_err_deg is None:
        return D

    beta_err = np.deg2rad(fwhm_err_deg)
    D_err = D * (beta_err / beta)
    return D, D_err
```

```
samples = {
    "seed": data_seed_corr,
    "F1": data_F1_corr,
    "F2": data_F2_corr,
    "F3": data_F3_corr,
    "F4": data_F4_corr,
    "F5": data_F5_corr,
}
```

```
### Scherrer analysis for all samples
def scherrer_for_all(samples, center=41.2, fit_window=2.5, bg_window=5.0, offset_step=2000, plot=True):
    rows = []
    ax = None

    if plot:
        fig, ax = plt.subplots(figsize=(7, 5), dpi=150)

    for i, (name, data) in enumerate(samples.items()):
        try:
            out = fwhm_at_peak_gaussian(
                data,
                center=center,
                fit_window=fit_window,
                bg_window=bg_window,
                plot=plot,
                offset=i * offset_step,
                label=name,
                ax=ax
            )
        except Exception:
            out = None

        if out is None:
            rows.append({
                "sample": name,
                "mu_deg": np.nan,
                "fwhm_deg": np.nan,
                "fwhm_err_deg": np.nan,
                "D_nm": np.nan,
                "D_err_nm": np.nan,
                "R2": np.nan
            })
            continue

        mu, fwhm, fwhm_err, r2 = out
        D_nm, D_err_nm = scherrer_size(fwhm, mu, fwhm_err)

        rows.append({
            "sample": name,
            "mu_deg": float(mu),
            "fwhm_deg": float(fwhm),
            "fwhm_err_deg": float(fwhm_err),
            "D_nm": float(D_nm),
            "D_err_nm": float(D_err_nm),
            "R2": float(r2)
        })

    if plot:
        plt.tight_layout()
        plt.show()

    return pd.DataFrame(rows)
```

```
df = scherrer_for_all(samples, center=41.3, plot=True)
df["D_with_err"] = df.apply(lambda row: f'{row["D_nm"]:.2f} ± {row["D_err_nm"]:.2f}', axis=1)
print(df)
```



	sample	mu_deg	fwhm_deg	fwhm_err_deg	D_nm	D_err_nm	R2	\
0	seed	41.291160	0.915238	0.023583	10.651618	0.274463	0.958127	
1	F1	41.250872	0.886201	0.023936	10.999163	0.297088	0.956871	
2	F2	41.273249	0.770653	0.021237	12.649254	0.348584	0.961466	
3	F3	41.231018	0.762383	0.026414	12.784707	0.442951	0.940667	
4	F4	41.239853	0.807940	0.025726	12.064161	0.384140	0.946193	
5	F5	41.274534	0.820857	0.023612	11.875679	0.341611	0.955002	